

Evolving Robotic Hand Morphology Through Grasping and Learning

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Abstract—Creatures can co-evolve their biological structures and behaviors under environmental pressures. Leveraging biomimetic evolution algorithms (referred to as co-design or co-optimization), a diverse range of robots with environmental adaptation has been generated. However, implementing these evolutionary methods or results in real-world robots, especially in the case of robotic hands, was not easy. In this context, this work presents a comprehensive self-optimization scheme for robotic hands that encompasses both software and hardware components. This scheme enables robots to autonomously refine their morphology through the integration of hardware gradients and reinforcement learning within parallel environments, thereby enhancing their adaptability to a variety of grasping tasks. For the hardware aspect, we developed a reconfigurable hand prototype with 37 variable hardware parameters (i.e., joint stiffness, the length of phalanges, finger location, and palm curvature) adjusted by mechanical components. Leveraging the adjustable hardware and 20 motors, this hand achieves full actuation and can dynamically adjust its morphology. The training results indicate that the fitness score of the self-optimizing hand exceeds that of original designs in this instance. The hardware parameters can be further fine-tuned in response to task variations. Moreover, the evolved hardware parameters are transferred to a real-world reconfigurable hand, demonstrating its grasping and adaptivity capabilities.

Index Terms—Evolutionary Robotics, Deep Learning in Grasping and Manipulation, Methods and Tools for Robot System Design, Hardware Gradient, Reconfigurable Hand.

I. INTRODUCTION

The design of robotic hands frequently emulates the properties of human hands, but it remains a challenging task to completely replicate human hand morphology, joint structure, actuation mechanisms, and other properties. Besides human-like design, we also can consider automatically evolving robots' design through the utilization of feedback from the grasping and manipulation tasks. This approach bears a resemblance to the evolutionary process of human hands.

Throughout human hand evolution, grasping and manipulating behaviors (e.g., grasping stones and using tools) are associated with numerous alterations in the morphology and

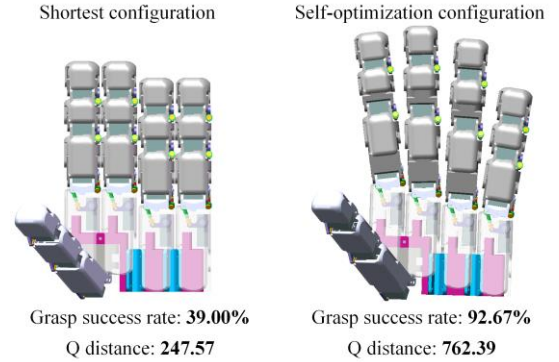


Fig. 1. Performance of the reconfigurable hand across different configurations. For a rigorous comparison, the hands employ a same policy model, differing only in their hardware configurations. The grasping success rate refers to the average success rate across a set of 20 objects, and the Q distance is also presented as an average value.

structure of the human hand. These changes include relative thumb length increases [1] and the differentiation of tendons [2]. The co-evolution of hand morphology and behavioral learning enhances both dexterity in manipulation and stability in grasping, a process biologically known as the Baldwin Effect [3]. Specifically, the two traits, morphology and behavior, are matched together in development and evolution to produce a unique functional phenotype [4]. Precision grip and manipulation potential [5] as a functional solution is a co-evolutionary consequence of human hand morphology and behavior, in which the natural environment plays a crucial role [4]. Despite the relative advantages of the human hand in manipulation potential [6], some studies speculate there is still room for further optimization by evaluating segmental models of different specimens [7].

The evolution process has provided valuable inspiration for the design of robots. Likewise, robotic hand morphology can be improved through grasping and learning, and made more adaptable to grasping objects or specific tasks. This approach fully leverages the hardware (body) intelligence and the software (brain) intelligence of robots, referred to as “co-design” [8], “co-optimization” [9] or “morphological computation” [10]. So far, this biomimetic approach has been applied in numerous fields, e.g., joint optimization of walking robots [11], co-design of robot arms [12] automated design of modular robots [13].

In this work, we proposed a highly efficient co-optimization scheme to acquire the optimal hand morphology (i.e., the length of phalanges, finger mounting location and palm curvature) and joint stiffness. The approach consists of two sequential optimization steps: control policy pre-training and variable hardware parameters gradient-based optimization. In the first

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step, the grasping policy was pre-trained using RL (reinforcement learning) while randomizing the variable hardware parameters to acquire “meta policy”. The “meta policy” is comprised of two components: the grasping policy and the stability model. In the second step, “meta policy” is fine-tuned (or rollout) in a 20-object set from the YCB benchmarks (Yale-CMU-Berkeley Dataset) [14]. The variable hardware parameters are optimized along gradients using the evaluation function $f(\delta)$ that is related to Q distance [15] (a force closure method) and the successful grasping quantity within the object set.

Moreover, we developed a reconfigurable fully-driven hand prototype with 37 variable hardware parameters and 20 motors. These variable hardware parameters are adjusted by mechanical components and include the length of phalanges, finger mounting location, palm curvature and joint stiffness. These motors drive MCP (metacarpophalangeal) joints, PIP (proximal interphalangeal) joints, DIPs (distal interphalangeal) joints and a CMC (carpometacarpal) joint.

This work has two core contributions, the first core contribution is to propose a gradient-based co-optimization scheme. Another core contribution is to develop a reconfigurable fully-driven hand prototype. Additionally, this work deploys the evolved hand morphology, demonstrating its grasping and adaptivity capabilities (Fig. 1).

II. RELATED WORK

In the field of co-optimization, a lot of previous seminal works have implemented optimization methods for the practical aspects of hardware design across a wide range of robot types. A graph heuristic search method is presented for co-designing over control and morphology through for single objective [16] and multiple objectives [17]. Besides MPC, RL is also used to generate robotic behaviors during robot design evolution [18]. The combination of RL and evolutionary algorithms is an intuitive approach inspired by human learning and evolution, which can be employed to optimize robot morphology [19], [20]. To improve efficiency, the robot morphology is optimized along gradients [11]. A few recent seminal works deployed their methods from simulation to the real world [21]. For example, quadrupedal robots are developed in the real world and can morphologically adapt to different environmental conditions [22], [23]. Besides walking robots, the real-world hardware evolution methods are also suitable for many other fields, e.g., soft robotics [24], robot arms [12], autonomous drones [25] and underactuated hands [26]. Leveraging the co-optimization approaches, effective hand designs naturally emerge to grasp diverse objects [27]. The soft hand morphology and control strategy are also automated co-design by grasping based on the SOFA framework [28]. For the real-world robotic hand, [29] proposes a mechanical and computational co-optimization scheme using RL. And [26] optimize the joint stiffness of the real-world underactuated hand using genetic algorithms and RL. [30] proposes an end-to-end differentiable framework and uses a novel parameterization for the continuous shape morphology based on cage-based deformation. [31] grow finger joints with different types and link lengths from null.

III. CO-OPTIMIZATION SCHEME

To achieve morphological evolution for environmental adaptation, we require two parts: morphological computation using the co-optimization method, and implementation using the reconfigurable hand.

A. Co-optimization Scheme

This scheme consists of two sequential optimization steps: “meta policy” pre-training and hand morphology gradient-based optimizing. In the first step, the grasping policy was pre-trained using RL while randomizing the variable hardware parameters to acquire “meta policy”. In the second step, the variable hardware parameters are gradient-based optimized in a set of objects while “meta policy” fine-tunes (or rollout). The purpose of hierarchical design is to reduce the coupling caused by policy search when updating hardware [11] (Fig. 2a).

In the first step, the “meta policy” comprises the grasping policy and stability prediction. The grasping policy generates a motor action sequence using RL. The stability model is used to terminate an episode when the grasping is predicted to be stable. The stability model is trained by the data from the buffer generated by the drop test. The data in the buffer is classified as a pair of adversarial samples, failure and successful samples. The episode will be terminated if $g(s_t) > 0.5$ (Fig. 2d).

In the second step, we construct an evaluation function $f(\delta)$ for gradient optimization. After downloading the shared hardware parameters, the agents roll out the “meta policy” which serves as the initial policy. And then, the rewards from all the local agents will be collected and accumulated as the label of the function $f(\delta)$. Random hardware parameters are pre-filled into the buffer, which facilitates the preliminary exploration of the entire design space. For stability, the variable hardware δ and the function $f(\delta)$ are fixed alternately during optimization, and the function $f(\delta)$ has a lower learning rate than δ . For exploration, the hardware parameters δ are sampled from a normal distribution, rather than acquired directly (Fig. 2b).

B. Fitness and Optimization

The reward function R consists of immediate rewards R_1 and final rewards R_2 .

$R_1 = \text{clip}(\lambda_1 \times (d_{closest} - \sum_{i=1}^{tips} \|pos_i - pos_{obj}\|), 0, 1)$. (1)
Where λ_1 is the weight, pos_i and pos_{obj} are used to compute the distance between all the fingertips and the grasped object. $d_{closest}$ is the closest distance in current episode. R_1 encourages the fingertips to approach the target, but this reward is not considered the final grasping metric. Consequently, we have assigned a minimal weight to λ_1 and stipulate that rewards are conferred only upon closer exploration within the proximity of the target. Furthermore, these rewards are constrained within the range of 0 to 1. Utilizing this metric as an immediate reward could mitigate the issue of sparse reward in RL. When the episode ends, the support for the grasped object is removed for the drop test. The final reward R_2 will be obtained if grasp succeeds in the test.

$$R_2 = \begin{cases} \text{clip}(\lambda_2 \times d_Q(o, H), 0, 10^3) & \text{if grasp succeeds} \\ 0 & \text{if grasp fails} \end{cases}. \quad (2)$$

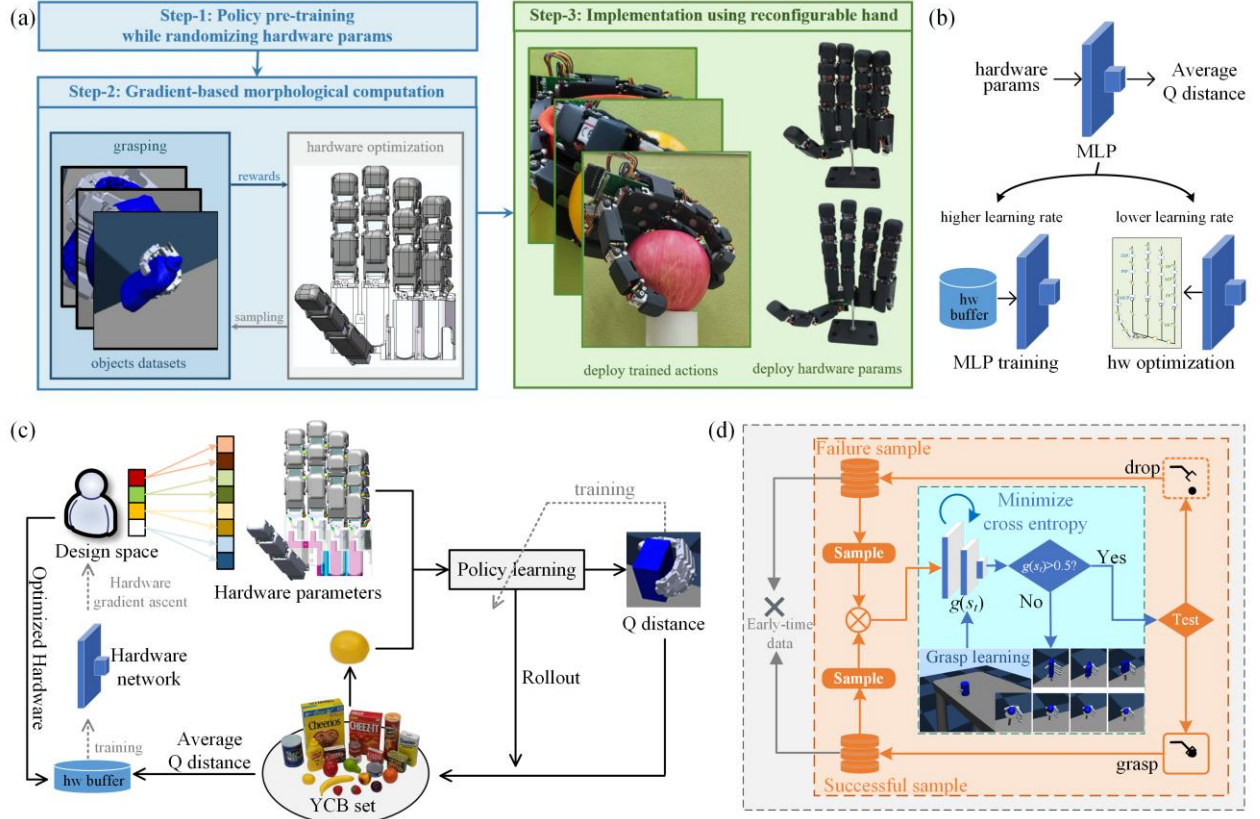


Fig. 2. Co-optimization scheme. (a) Co-optimization scheme presents the three key steps: the “meta policy” training, gradient-based morphological computation and transferring to reconfiguration hands. (b) presents the training and utilization of hardware network. (c) presents the self-optimization process of the reconfigurable hand through interaction experience with multiple objects. The hardware parameters are generated by the hardware network and tested by grasping in parallel environments. The average Q distance calculated after the grasping test is added to the hardware buffer. The hardware network is trained by the hardware (hw) buffer and provides optimization direction along the gradient for the current design parameters. The design parameters do not necessarily correspond exactly to the adjustable hardware parameters; when one design parameter corresponds to multiple hardware parameters, it indicates that these hardware parameters are coupled. We selected 20 items from YCB benchmarks, rather than all of them. Similar objects in shape and size were excluded, as were objects that are not suitable for single-hand grasping, such as the mini soccer ball. (d) Grasp stability model is a three-layer MLP, the cross entropy of which is minimized by successful and failure samples from drop test.

Where λ_2 is the weight, that has been chosen through numerous tests. The value of 10^4 from the study [32] is also available for R_2 . $d_Q(o, H)$ is the Q distance from origin o to convex hull H , which is frequently as a metric of grasping quality.

$$d_Q(o, H) = \begin{cases} \min \| \sum_{z_1}^I (z_i \times w_i) \| & \text{if } o \text{ out } H \\ -\min_{h \in P} (\min \| \sum_{z_1}^I (z_i \times w_{h,i}) \|) & \text{if } o \text{ in } H \end{cases} \quad (3)$$

When o is outside H , the grasping is considered unstable, and its instability increases with greater $d_Q(o, H)$. The shortest distance is equivalent to solving the following quadratic programming problem. The coefficient $z_i > 0$, and $\sum z_i = 1$. I is the quantity of convex hull vertices, and w_i is the vertex coordinates in the wrench space. When o is inside H , it is considered stable, and its stability increases with greater $d_Q(o, H)$. the convex hull is decomposed into multiple hyperplanes P to find the hyperplane closest to the origin. The cost of this computation is higher than the last situation.

For a set of multiple objects, the total fitness $F(\delta)$ is

$$F(\delta) = \sum_{\pi_0}^{\pi_n} R_2. \quad (4)$$

The fitness function $F(\delta)$ does not directly optimize the hardware parameters but optimizes the evaluation function $f(\delta)$ of hardware parameters through MSE (Mean Squared Error).

$$\text{MSE} = \frac{1}{k} \sum_{i=1}^k (F(\delta_i) - f(\delta_i))^2. \quad (5)$$

Where k is the batch size sampled from the hardware buffer. For the hand that is interacting online, its hardware parameters are updated using gradient ascent (Fig. 2c).

$$\delta_{\text{new}} = \delta_{\text{old}} + \alpha \nabla f(\delta). \quad (6)$$

The updated hardware parameters δ_{new} will be appended to the hardware buffer for further training.

C. Hardware Self-Optimization

Overall, the specific method for the self-optimization process of hardware utilizes the idea of the Actor-Critic in reinforcement learning. Leveraging the experiences from environmental interaction, the evaluation function of the hardware is trained, and the hardware parameters are optimized through gradient ascent. We use joint stiffness optimization as an example to illustrate the specific method. Firstly, the hand self-optimization runs on 20 CPU cores via the gym library and MuJoCo [28], with each core having an independent grasping environment for the specified object. After all environments reset, the joint stiffness is adjusted by the joint parameter “springref” on the four-bar linkage of the MJCF file. After that, we use Soft Actor-Critic [34] to train a pre-trained model under

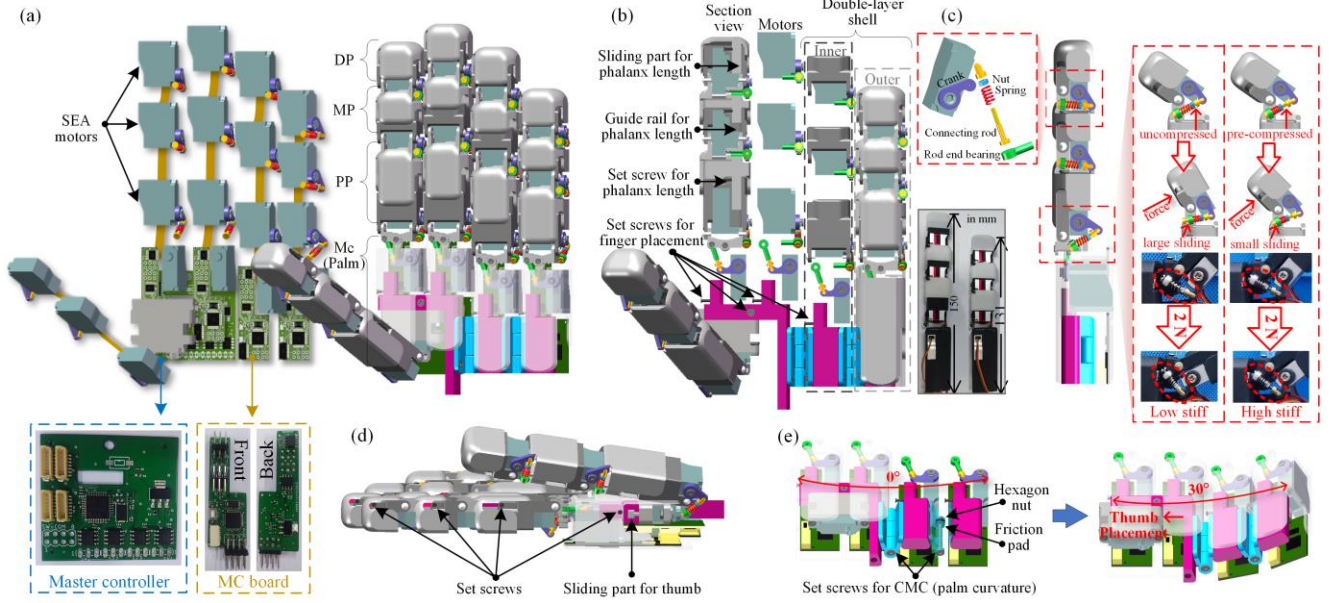


Fig. 3. The reconfigurable hand and its adjustable hardware. (a) presents the hand appearance and embedded control system. The master controller is embedded in the palm; the MC (metacarpal) boards are embedded in metacarpals of the five fingers. Mc is metacarpal, PP is proximal phalanx, MP is middle phalanx, DP is distal phalanx. (b) illustrates the internal structure of the reconfigurable hand. The little and ring fingers respectively display the outer and inner shells of the phalanges; this dual-layer shell design facilitates phalanx length adjustment. The middle finger exhibits the motor distribution, where the MCP swing joint is actuated by a four-bar linkage mechanism, and the other three flexion joints are driven by SEAs with a similar four-bar linkage mechanism. The index finger provides a section view, with phalanx length adjustment comprising a pair of fixable sliding parts, including a sliding part, guide rail, and set screw. Finger placement has a similar adjustment mechanism, with set screws and the Mc shell moving along the red bracket. The Mc shell is presented in (a). (c) The stiffness of the SEA is adjustable via springs. The enlarged view details the joint stiffness adjustment mechanism, utilizing a nut to modify the spring pre-compression, thereby altering the antagonistic force. A smaller pre-compression will cause the connecting rod to produce a larger passive sliding under force, which will exhibit stronger adaptability. The completed experimental images will also be presented in the joint stiffness experiment. (d) offers a lateral view of the hand, clearly showing the positions of the set screws and the structure of the thumb's sliding part. (e) depicts the palm, demonstrating the rotation of the ring finger and little finger CMC joints (palm curvature). The palm curvature can be modified using pairs of fixable rotation parts, consisting of a hex nut, friction block, set screw, and hinge. The hex nut is constrained from rotating by the hinge shell, and tightening the set screw increases hinge friction for locking. The movement of the thumb is also presented.

randomized joint stiffness. These data will be pre-filled into the hardware buffer. Subsequently, the hardware network will be updated by the hardware buffer and provide the direction of the gradient for joint stiffness optimization. During training, the grasping strategy will also be continually fine-tuned based on the feedback of the Q distance from different environments. Reinforcement learning, the hardware network, and gradient computation are all based on the PyTorch framework.

IV. EXPERIMENTS

This section will be divided into three parts: reconfigurable hand design, experimental setup, and results.

A. Reconfigurable Hand Design

The reconfigurability of the hand enables it to dynamically adjust its morphology and joint stiffness based on the environments and grasped objects. This mechanical intelligence makes trade-offs across multiple objects by taking both a globally optimal design and control outputs into consideration.

The hand has 37 variable hardware parameters (Tab. 1). For the hand morphology, we utilize a pair of fixable sliding parts to adjust the length of the phalanges and the mounting location of fingers. The orientation of the little and ring fingers can be adjusted by a pair of fixable rotating parts. These two fixable

TABLE I
RECONFIGURABLE HARDWARE PARAMETERS

	Total Number	Distribution		
		Thumb	Finger	Palm
Joint stiffness	15	3	3x4	0
Phalanx length	15	3	3x4	0
Finger placement	5	1	1x4	0
Palm curvature	2	0		2

TABLE II
DISTRIBUTION OF 20 MOTORS

	CMC	MCP	PIP	DIP
Thumb	2	1		1
Non-pollical	/	2	1	1

joints can be considered as the CMC joints of the ring and little fingers, which adjust the curvature of the palm of the cross-section. For the joint stiffness, the SEAs (series elastic actuators) can adjust stiffness in 15 joints. SEA's motor is a servo, model KST X06. The SEA is a four-bar linkage mechanism with an elastic component mounted on the connecting rod, and the movement of the output link on the connecting rod is affected by the adjustable antagonistic resistance of the springs (Fig. 3).

This reconfigurable hand is capable of being fully actuated by 20 motors. Specifically, the thumb's CMC joint is actuated by 2 motors, the MCP joint by 1 motor, and the IP (interphalangeal)

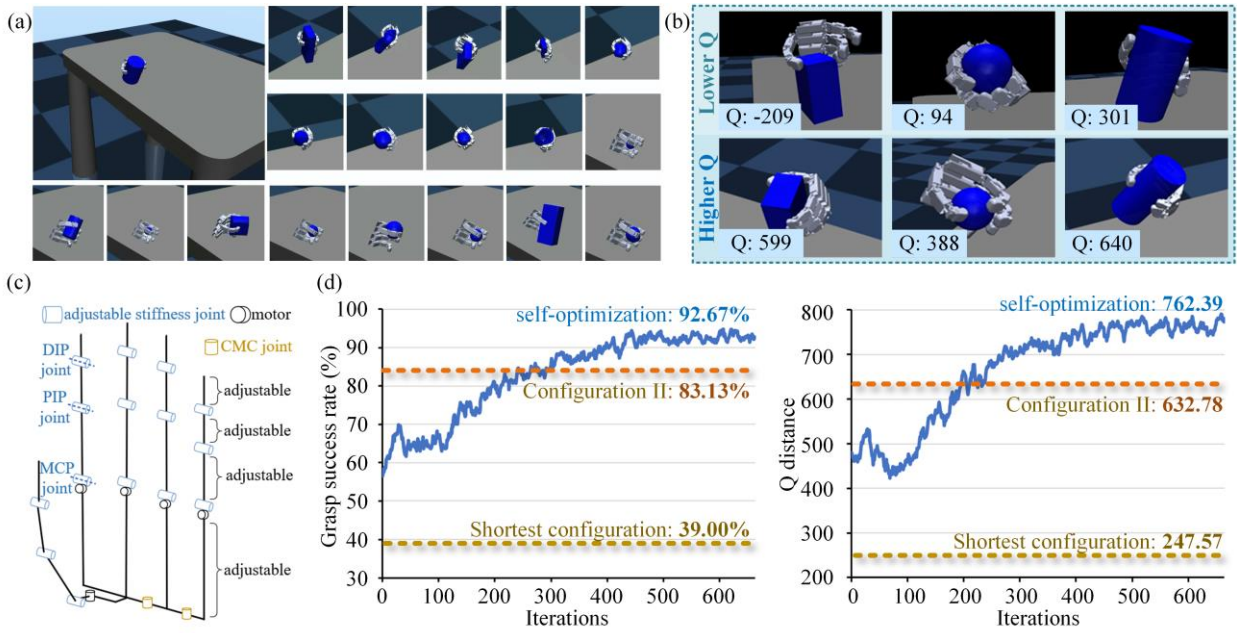


Fig. 4. The training of hardware. (a) This presents the training platform in parallel environments. The complete grasping demonstration can be found in the attached video. (b) Different grasps exhibit different force closure values (Q distance), despite achieving successful grasps. Consequently, this work adopts the force closure method (Q distance) as the metric, due to its capacity to augment the quality of the grasp, rather than only the grasp success rate. (c) This schematic diagram presents the distribution of training hardware. The yellow cylinders are angle-adjusting parts, the black cylinders are normal motors, and the blue cylinders are joints with SEAs. (d) The performance of the shortest configuration is configuration I, while configuration II is the reconfigurable hand with the phalanx lengths referencing a hand bone specimen. For a rigorous comparison, all three configurations employ the same pre-trained model, and the pre-trained model is not fine-tuned during hardware network training to ensure that improvements in performance were solely attributable to morphological changes.

joint by 1 motor. For the non-pollical fingers, the MCP joint is actuated by 2 motors, the PIP joint by 1 motor, and the DIP joint by 1 motor (Tab. 2). Flexion joints are actuated using SEA to maintain elasticity during grasping, and swing joints use the general motor to reduce the hand size.

The hand circuit adopts an architecture of one master and four slaves. The master PCB (Printed Circuit Board) is mounted on the back of the thumb, and communicates with the other four fingers through SPI. The four-finger PCBs have the STM32 as the MCU (Microcontroller Unit) and are mounted inside the MC (metacarpal) of each finger. Each MC board controls the four motors using PWM and reads the current from the four motors to estimate joint torque. The master PCB collects and uploads this sensing information using a serial port and distributes position control commands to each finger, and control thumbs.

The circuit and structure of the finger are modular in design: the finger can be used after being inserted and fixed in the corresponding location. By adjusting the configuration of the fingers, different fingers can substitute for each other, and both the differences and interchangeability can be met.

B. Experimental Setup

As mentioned above, we have developed a simulated hand in MuJoCo that corresponds to a real-world reconfigurable hand. In addition to the 20 motors, 6-DoF actuators have been added to control the position and orientation of the entire hand. The feedback state includes the hand joint positions, joint torques, action feedforward, hand position and orientation relative to the object, object position and orientation, shape primitives and sizes of the grasped object, and design parameters. The grasp

planning for different hand morphologies is not specified by humans but is trained through RL based on the aforementioned feedback states, thereby minimizing human interference. Furthermore, for a rigorous comparison, we utilized the identical trained RL model, which was not further fine-tuned during morphological evolution. This ensures that the Q distance is solely attributed to hardware adjustments. Specifically, the object is initially placed on a support before the grasping process. The hand executes the action from the RL model and subsequently transfers the next state back to RL. When the stability model predicts stability, the motors maintain the current position control actions. Thereafter, the support is removed, and a drop test is conducted.

The touch sensor on the desktop is used to detect the result of the drop test. If the touch sensor on the desktop detects contact for a certain period after the episode ends, the object is considered to have been dropped and bumped onto the table. Conversely, if contact is detected in the hand, rather than on the desk, the object is considered grasped (Fig. 4a).

Rather than directly optimizing the 37 parameters, we optimize them through the design space. This approach implies that certain parameters are coupled. The design space includes one joint stiffness parameter, five finger length parameters, one thumb location parameter, and one palm curvature parameter. There are two reasons for this approach: Firstly, humans can impose constraints on a robot's morphology for aesthetic or design purposes, e.g., the proportions of the phalanges. Furthermore, the optimization outcomes remain satisfactory even under constraints. Secondly, thorough exploration of high-dimensional space is difficult. Imagine that sampling once from

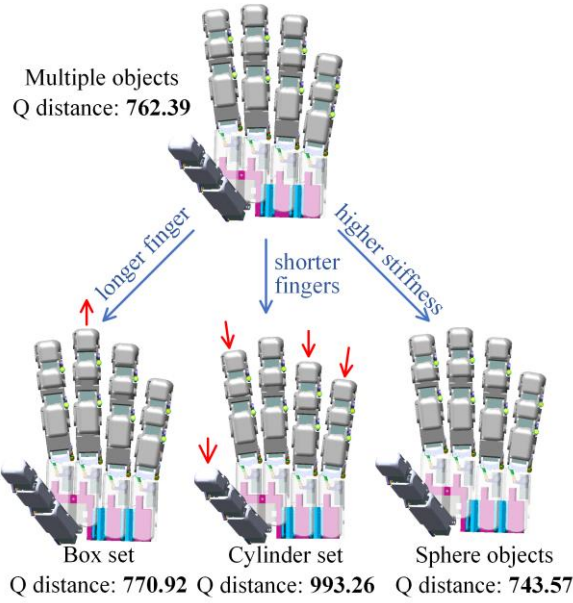


Fig. 5. Morphological fine-tune under diverse object Sets. The fine-tuning direction for hardware parameters exhibits variation for different object sets.

a half subspace of all dimensions would require 2^{37} .

C. Results

Robot's Self-optimization: we test the performance of the three configurations of a reconfigurable robotic hand by using the same RL pre-trained model to ascertain their mean grasping success rates and average Q distances across the same set of objects (Fig. 4d). In comparison to configuration I, configuration II demonstrated a significant enhancement in grasping success rate, escalating from 39% to 83.13%, and an improvement in the average Q distance from 247.57 to 632.78. Nonetheless, there remains potential for further enhancements in performance. To explore this potential, we use the hardware evaluation network that enables the robotic hand to autonomously optimize its hardware parameters in alignment with the gradient direction. For a rigorous comparison, we utilized the identical pre-trained reinforcement learning model employed in configurations I and II, and ceased further fine-tuning. It ensures that improvements in performance were solely attributable to morphological enhancements. The outcomes of this self-optimization process revealed that the robotic hand's average grasping success rate escalated to 92.67%, surpassing both configuration II's 83.13% and configuration I's 39%. Furthermore, the average Q distance achieved was 762.39, exceeding the figures recorded for both 632.78 of configuration II and 247.57 of configuration I (Fig. 4).

The optimized hardware parameters are listed (Tab. 3). The palm curvature has been reduced due to the absence of soft tissue and elastic structures. A flatter palm facilitates the grasping of cylindrical objects. The pre-compressed length of the spring was minimized to zero to fully leverage its adaptability. Additionally, the optimized lengths for the thumb, index finger, middle finger, ring finger, and little finger are 102.9 mm, 165.1 mm, 169.5 mm, 164.3 mm, and 133.8 mm, respectively. Furthermore, previous research [36] indicates that the average human finger lengths and

TABLE III
THE SELF-OPTIMIZED HARDWARE

	F1	F2	F3	F4	F5
Metacarpal	46.8	73.0	67.9	61.8	56.6
Proximal Phalanx	32.6	45.2	48.7	47.6	35.5
Middle Phalanx	/	27.9	32.7	33.3	22.7
Distal Phalanx	23.5	19.0	20.2	21.6	19.0
Palm Curvature	/	/	/	0.3°	0.3°
Pre-compression	0 mm				

* F1~5 are the thumb, index, middle, ring, and little fingers, respectively. The phalanx lengths are in mm, metacarpal length reflects finger placement.



Fig. 6. Reconfigurable grasping demonstration. The hardware is based on morphological computation and reconfiguration.

their standard deviations are 99.46 ± 5.28 mm, 146.10 ± 8.67 mm, 152.96 ± 7.48 mm, 142.32 ± 7.50 mm, and 120.50 ± 6.26 mm, from thumb to little finger. Compared to human hands, the optimized reconfigurable hand exhibits longer finger lengths, except for the thumb. This discrepancy may be attributed to the structural differences between reconfigurable hands and human hands. Overall, we believe that integrating insights from human hand evolution with robotic self-optimization could be highly significant for future research.

Morphological Fine-tune under Diverse Object Sets: These 20 objects are divided into three sets: cylinders, boxes, and spheres, although alternative classification methods, such as size or randomness, can also be applied. The hardware in the three tasks is independently trained, with an average of 835.92, surpassing the original larger set's average of 762.39. This enables the reconfigurable hand to achieve higher performance within smaller object sets (Fig. 5). However, this also means that the fine-tuned hand may exhibit weaker adaptability to other grasping tasks. Therefore, we further performed cross-validation on these three sets of objects with the other two fine-tuned morphologies. For the box set, the Q distances of the other two morphologies are 676.1 and 755.0, which are lower than the previous 771. For the cylinder set, the other two Q distances are 995.1 and 995.2, which are close to the previous 993.26. For the sphere set, the other two Q distances are 681.6 and 701.0, which are also lower than the previous 743.57.

Hardware Configuration Transfer to Real-world Hand: The transfer of control policy requires corresponding actuators for real-world robots. Similarly, the transfer of trained hardware necessitates the availability of adjustable mechanical components in the real world. These components are presented in Fig. 4. In terms of grasping policy, rather than directly deploying it onto the reconfigurable hand, we input the corresponding shape primitives and sizes of real-world objects

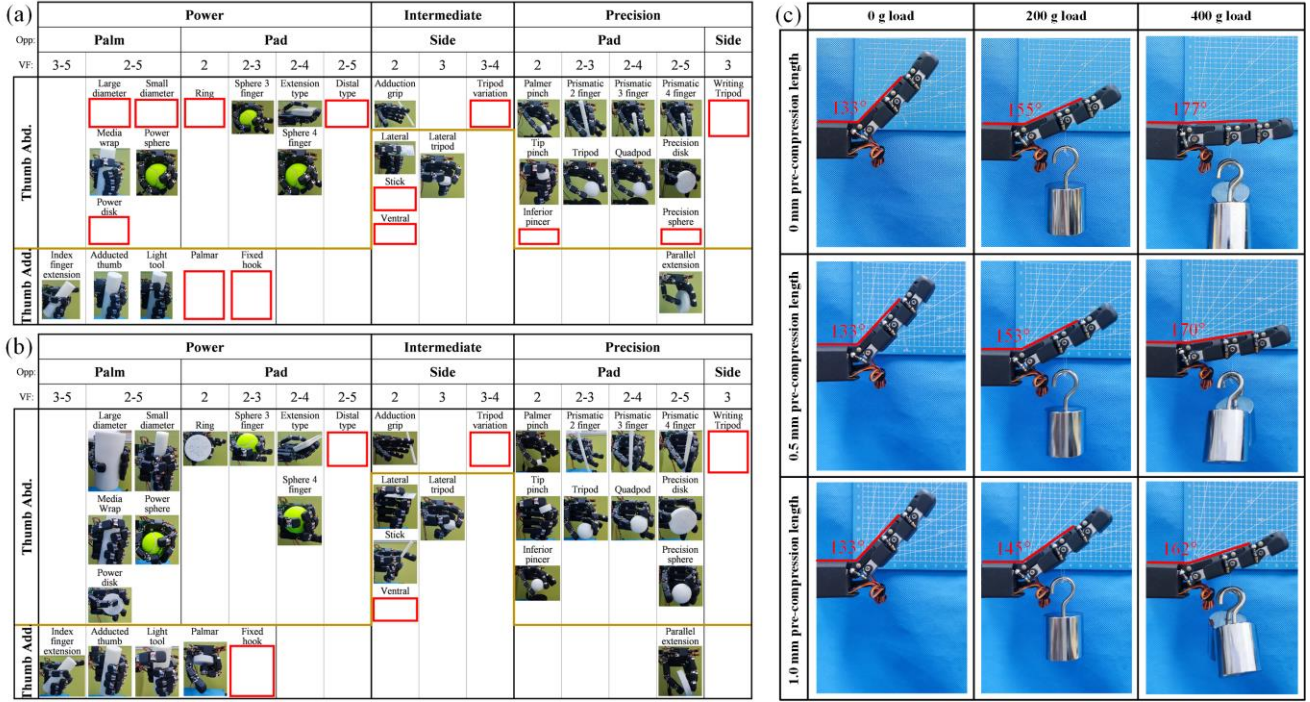


Fig. 7. Test of grasping and adaptivity of the reconfigurable hand. (a-b) The hand can complete 20 types of daily grasps in Feix taxonomy before self-optimization. After self-optimization and reconfiguration, the number of grasping types increases to 28. (c) The hand joint adaptability can be adjusted by the pre-compression length of the SEA spring. When the pre-compression length is 0, a 400-gram load can produce 44° of passive motion; when the pre-compression length is 0.5 mm, it decreases to 37°; with a pre-compression of 1.0 mm, it decreases to 29°.

into the trained model to generate motion actions. The final actions are then sent to the main MCU of the reconfigurable hand, resulting in the generation of real-world grasping behaviors (Fig. 6).

Test of Grasping and Adaptivity: The hand evolution requires trading off multiple objects and self-optimizing towards a common advantage. To compare the generalization ability of the reconfigurable hand to different objects before and after evolution, we used Feix taxonomy [35] as the testing condition. Feix taxonomy includes 33 types of grasps, nearly covering all human grasps in daily life. The evolved hand can perform 85% (i.e., 28/33) of these grasps, while the pre-evolved hand can perform 61% (i.e., 20/33). The evolved hand exhibits a significant increase in phalangeal length. This change helps in grasping larger objects (e.g., large diameter, ring), and the change of phalangeal ratio also enhances its ability to grasp objects requiring smaller form closure (e.g., small diameter, palmar). In addition, the optimization of joint stiffness also significantly helps improve the grasping ability (e.g., inferior pincer, power disk) (Fig. 7a).

Furthermore, to demonstrate the adaptivity of SEA, we tested the passive motion capacity of the MCP flexion joint under pre-compression lengths of 0, 0.5, and 1.0 mm. Pre-compression is the pre-loading of the SEA spring using the nut to increase the antagonistic force of linkage slippage, as illustrated in the enlarged diagram in Fig. 3c. The unloaded (0 g load) joint angle is set at 133°. Each of these SEA configurations is subjected to 200 g and 400 g weights. The passive angular displacements for 0 mm pre-compression length are 22° and 44°; that of 0.5 mm pre-compression length is 20° and 37°; that of 1.0 mm pre-

compression length is 12° and 29°. As the pre-compression length increases, the distance of linkage slippage of SEA decreases, and the range of passive motion gradually decreases (Fig. 7b).

V. CONCLUSION AND FUTURE WORK

We proposed a complete scheme for morphological computation of the reconfigurable hand, which is capable of dynamically adapting to environments through morphology evolution and grasping learning. This scheme includes software and hardware: the gradient-based morphological computation method and the reconfigurable hand prototype with adjustable morphological and mechanical parameters. Based on this scheme, we transferred the computational results of hardware network to the real-world reconfigurable hand.

This idea is inspired by the evolution of human hands in nature and endows the robotic hand with the ability of self-optimization, demonstrating how reconfigurable hands evolve towards different morphologies under diverse object sets. This idea of co-evolution of morphology and control is related to the concept of the Baldwin Effect. Nevertheless, this work does not delve deeply into the evolution of human hands due to the disparity between reconfigurable hands and actual human hands. Therefore, in our future work, an important direction will involve applying this computational process to the human musculoskeletal model. This approach aims to validate the manipulative advantages of human hands over those of other primates [6] and to explore if there is further evolutionary room for improvement [7].

The current scheme has some limitations. In terms of

hardware, hand development has high integration demands in the electromechanical system due to space limitations. However, reconfigurability (e.g., adjustable finger length and adjustable stiffness) requires additional space. This increases the hardware design difficulty. In terms of software, the computational resources increase rapidly with the number of hardware parameters for self-optimization. This extends the time to resolve these hardware parameters. Furthermore, the topology of the robot (e.g., the number of joints or fingers) cannot be altered within our current framework, as robots with different variable hardware parameters share a common action space and physical characteristics. This limits the transferability across robotic hands with different fingers. In future work, we will add actuators to the adjustable mechanical parts for the reconfigurable hand and study a more efficient co-optimization method to achieve environmentally adaptable online co-optimization for the real-world hand. Additionally, we will explore incorporating Graph Neural Networks into the self-optimization method.

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